

ARK-483 — Preregistration

Verification Decision Latency

Series: ExecutionProof authorization-boundary corpus — **Production-Boundary phase (P01)**

Experiment ID: ARK-483

Institution: Remnant Fieldworks Inc.

PI: Derek Adam Hone

Substrate: Classical software (no quantum hardware, no cryptographic security claims)

Benches: Dual independent — V1 (JavaScript), V2 (Python)

Component Under Test (CUT): the ARK-458 exact-action-binding guard (`evaluate()`)

Preregistration status: LOCKED before official execution (see `MANIFEST.txt`)

Commercial / standards / IP justification (required by RF covenant)

- **Commercial purpose:** Directly answers **Prospect Question #2** — “What is the verification latency?” — the single most-asked buyer question that the RF corpus had **never measured**. Every prior experiment established whether ExecutionProof decides correctly; none had established how fast a decision is. This is the first RF latency measurement, and it measures the real production-boundary decision path built in ARK-458.
 - **RF-100 requirement:** Provides the quantitative basis for a future **verification-overhead budget** clause — the claim that gate verification adds negligible latency to an approved action.
 - **Patent-family connection:** Produces dated **working-example** evidence that the exact-action verification mechanism (filed parent `19/529,283` and the AI-governance / proof CIP family) is computationally cheap enough to sit inline on an authorization path. Working-example evidence only — it does not legally validate any claim, and adds no new matter to any filed application.
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Question

How long does a single ExecutionProof **verification decision** take? Specifically, for the ARK-458 exact-action-binding guard, what is the per-decision wall-clock latency (mean and tail percentiles) for the ALLOW path (exact match, all dimensions compared) and the DENY paths (mismatch), on the reported reference machine — and is it low enough to sit inline on a production authorization path without a meaningful latency penalty?

Component Under Test

The **frozen** ARK-458 guard `evaluate()` — measured directly, in-process, with no network, no I/O, no process spawn between calls. ARK-458 established this guard’s correctness (verdict PASS, dual-guard concordance 800/800). ARK-483 measures only its speed. The ARK-458 guard files are locked and are **not modified**:

- **V2 (Python):** `ark-458/verifiers/v2_guard.py :: evaluate` — imported directly.
- **V1 (JavaScript):** `ark-458/verifiers/v1_guard.js :: evaluate` — loaded as its exact **locked bytes** into a `vm` sandbox (its `stdin main()` loop is not executed). No edit is made to the locked file; the bytes measured are the frozen guard.

Design

Three decision paths are measured **independently in each implementation** (V1 and V2):

Path	Description	Decision	Work performed
<code>allow_exact_match</code>	execution action equals approved action on all 5 dimensions	ALLOW	all 5 dimensions compared
<code>deny_first_dim_mismatch</code>	first binding dimension (principal) differs	DENY	early exit on dimension 1
<code>deny_last_dim_mismatch</code>	last binding dimension (condition) differs	DENY	worst-case DENY — all 5 compared before mismatch

- **Warmup:** 5,000 discarded iterations per implementation (JIT / cache warm).
- **Cold start:** the first-ever decision latency is recorded separately (not folded into the warm sample).
- **Timed sample:** $N = 100,000$ timed iterations **per path**, each timed individually with a high-resolution monotonic clock (`time.perf_counter_ns` in Python, `process.hrtime.bigint()` in Node).
- **Reported statistics per path:** mean, median, p50, p95, p99, p99.9, min, max, stdev, throughput (decisions/sec).
- **Environment** (CPU model, language/runtime version, OS) is captured into the results file so every absolute number is bound to the machine that produced it.

Metrics

- `p95_us` per path = 95th-percentile per-decision latency in microseconds.
- `worst_path_p95_us` = max p95 across the three paths (the figure judged against the ceiling).
- Secondary: cross-implementation concordance — V1 and V2 worst-path p95 should fall in the **same order of magnitude** (both are independent implementations of the identical procedure; a large divergence would indicate a harness artefact).

Pass criterion (fixed before execution)

- **C1 (latency ceiling):** `worst_path_p95_us` $\leq 1000 \mu\text{s}$ (1 ms) for a single in-process decision, **in each implementation independently**.
- **Verdict = PASS** iff C1 holds for both V1 and V2; else FAIL. The measured numbers are published either way.

The 1 ms ceiling is a deliberately generous, buyer-legible bar (“a verification decision adds under a millisecond”). It is preregistered so the result cannot be reverse-fit; the honest expectation is that the true figure is far below it, and the published numbers — not the ceiling — are the deliverable.

Kill conditions / validity gate

1. **Decision-sanity gate:** before timing each path, the CUT is invoked once and its decision **must** equal the path’s expected decision (ALLOW/DENY). If any path does not produce its expected decision, the run **ABORTS** (no verdict) — a latency number for the wrong decision is meaningless.

2. **Non-triviality:** $N \geq 100,000$ timed iterations per path and warmup $\geq 5,000$, fixed before execution, so the sample is not a micro-artefact.

Publication rule

The outcome (PASS, FAIL, or ABORT) will be published to GitHub with the raw per-path results JSON for both implementations, and defensively published to Zenodo (CC BY 4.0), regardless of verdict.

Honest bounds

- This measures the **in-process decision only**. It explicitly **excludes** network round-trips, authorization-token retrieval, evidence/ProofRecord persistence, logging, serialization at a service boundary, and process startup. A real deployed gate will be slower; this is the **floor**, the cost of the decision logic itself.
- Absolute latencies are **bound to the reference machine** reported in the results file. They are not a guarantee on any other hardware, runtime, or load condition.
- This is a micro-benchmark of **control logic**, not a load test, not a concurrency/throughput test under contention, and not a production SLA.
- No claim that this experiment legally validates any patent claim or certifies RF-100 conformance. Working-example evidence only.

Preregistered under the Remnant Fieldworks Standing Covenant (preregister → lock → execute → publish all outcomes). To God be the glory. Proof Before Power. Verification Before Execution.